

Suvanth Erranki

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EDUCATION

Carnegie Mellon University – GPA: 4.0/4.0

Bachelor of Science, Computer Science

Expected May 2029

Pittsburgh, PA

Relevant Coursework: Matrix Theory, Imperative Computation, Math Foundations for CS, Putnam Seminar

Princeton University – GPA: 4.0/4.0

Dual Enrollment, Mathematics

May 2026

Princeton, NJ

Relevant Coursework: Real Analysis, Number Theory

ACTIVITIES & EXPERIENCE

Software Engineering Intern, Lockheed Martin

Jun. 2026 – Aug. 2026

- Engineered an end-to-end aerospace cost-analysis pipeline in Python, loading a 20-column, 220K row dataset of Aircraft models and parts into an interactive Plotly/Dash dashboard.
- Constructed cost-flow graphs with Kahn's topological sort over 185K nodes and 169K edges, flagging 15K+ problem nodes.
- Developed an NL-to-SQL pipeline that generates validated DuckDB queries from aerospace prompts using the internal Genesis LLM API. Implemented a Pandas query executor to filter 220K rows in 22ms.

Lead Organizer, West Windsor-Plainsboro Programming Competitions

Feb. 2024 – Jun. 2026

- 501(c)(3) sponsored by Jane Street, HRT, and more; hosted 3 programming competitions for 500+ students (latest at TCNJ); distributed \$10K+ in cash prizes, raffles, and merchandise.

AWARDS

USA Computing Olympiad Platinum & US Open Championship Invitee (1 of 75)

Mar. 2026

MIT Battlecode 2025 Finalist – 1st/113 HS Teams, 7th/504 Overall ([Finals Broadcast](#))

Jan. 2025

1st Place — Lockheed Martin Code Quest Advanced (30 teams)

Feb. 2026

1st Place — Cornell HS Programming Contest (75 teams)

Mar. 2025

1st Place — UPenn PClassic Advanced (50 teams)

Apr. 2025

PROJECTS

WWP Programming Competitions Online Judge | [TypeScript](#), [Node.js](#), [Linux](#)

- Built a secure multi-language online judge in TypeScript/Node.js supporting C/C++, Java, Python, and PyPy with Linux `isolate` sandboxes, custom checkers, testcase generators, and automated scoring.
- Implemented a fault-tolerant distributed grader with authenticated job scheduling and automatic recovery from interrupted tasks, evaluating 5K+ submissions from 500+ users across 3 contests.

Chess Engine “Kinetic Hero” | [C++23](#) | [Source Code](#)

- C++ chess engine incorporating Monte Carlo Tree Search (MCTS) and bitboard-based move generation.
- Optimized move generation with precomputed attack tables and magic-bitboards, yielding 60M nodes/sec on a single Apple M2 Pro core.

SKILLS

Languages: C++, Python, C, Java, JavaScript, TypeScript, HTML/CSS

Systems & Tools: Linux, Docker, Git, PostgreSQL, MySQL, Bash, Vim

ML/Data: PyTorch, NumPy, pandas, Matplotlib

Frameworks: React, Node.js, Vue